



Team Aaloo: The Smart Solar Lamp Revolutionizing Fish Farming

Sector: Sustainable Aquaculture | Renewable Energy | IoT & Smart Agriculture

Category: COMPUTING | AI & DIGITAL INNOVATION.



Healthy Fish, Healthy Communities



1

Critical Protein Source

Over 60% of all animal protein consumed in Bangladesh comes directly from fish, highlighting its irreplaceable role in the national diet and public health.

2

Massive Livelihood Support

More than **2+ million people** depend on fisheries for their livelihoods, encompassing fishermen, traders, processors, and related industries.

Beyond the Plate: When Fish Health Becomes a Human Risk



Heavy Metals

- **Safe limit:** Mercury $\leq 0.5 \text{ mg/kg}$ in fish (EU/FAO standard).
- **Current situation:** Lead and cadmium **bioaccumulate** in farmed fish over time, especially from contaminated feed and water.

Source: FAO; NIH (2022)



Microplastics

Contamination: Fishmeal contains 0–526 particles/kg,

Current scale: 5–12 million tons of microplastics enter oceans each year.

Risk: Microplastics (0.15–5 mm) enter fish via feed and water, posing food safety concerns.

Source: (2021) Environmental Toxicology and Chemistry Review



Chemical Residues

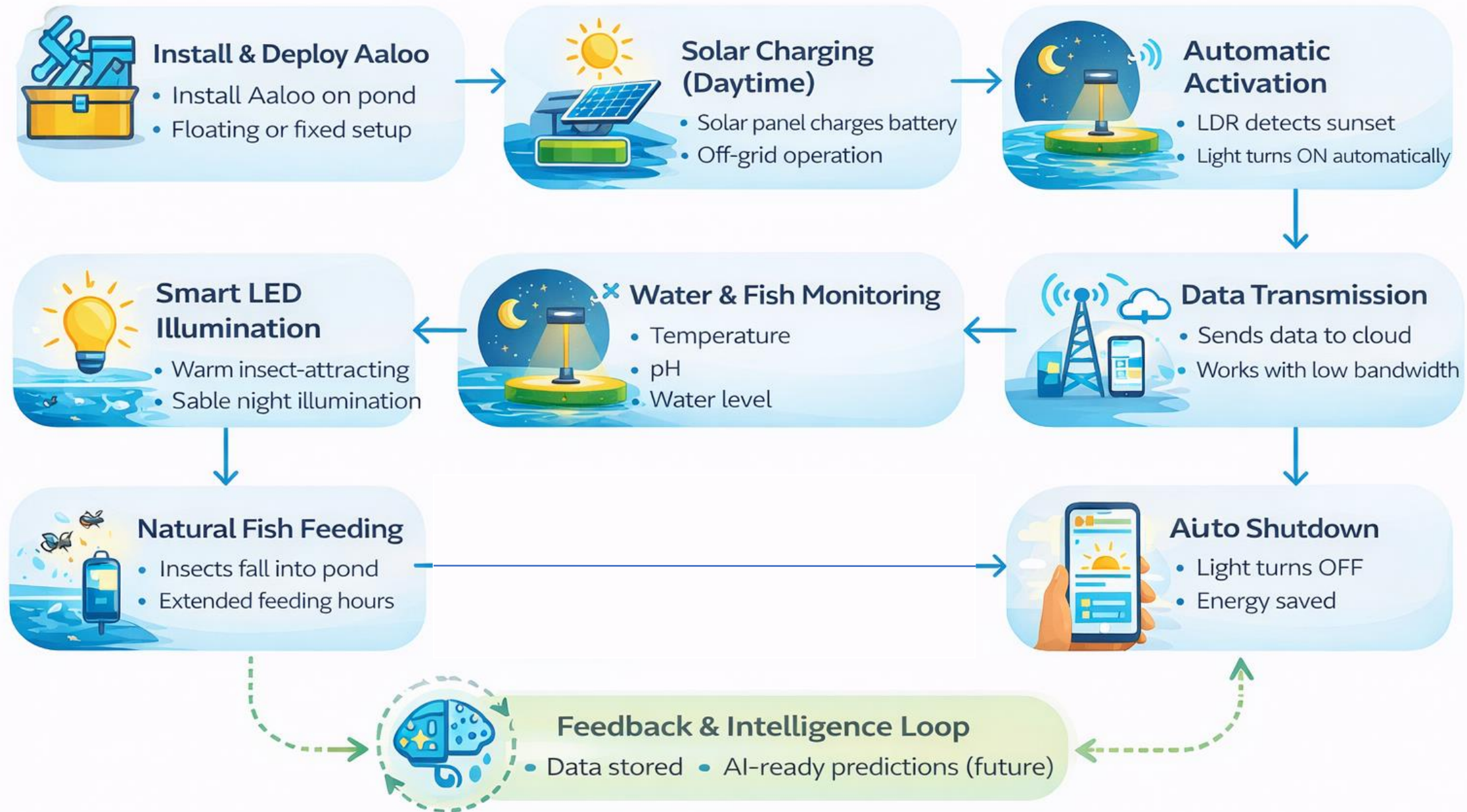
Pesticides, PCBs, and dioxins accumulate in farmed fish. EPA tolerable dioxin limit: **0.7 picograms/kg body weight/day**. Fish consumption alone can exceed safe daily limits.

Source: EPA Guidelines; Nutrition Facts Research (2025)

Aaloo



HOW AALOO WORKS

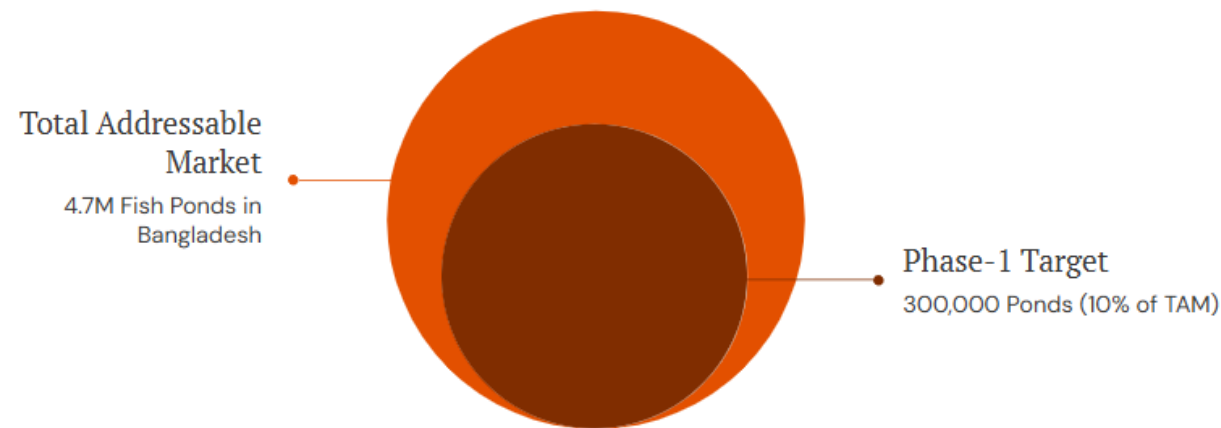


Local Market Size – Bangladesh



4.7 Million Fish Ponds

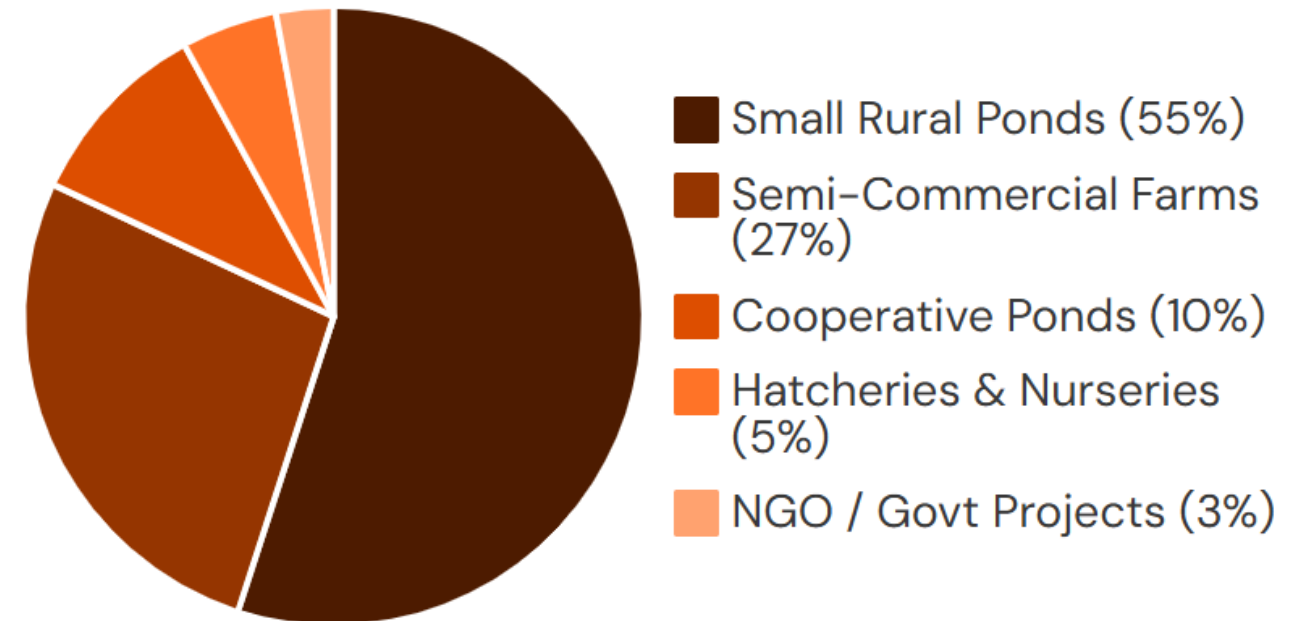
Total Addressable Market (TAM)



Focused on ponds using night lighting and suitable for solar floating systems

Data Source: Department of Fisheries (DoF), Bangladesh

Target Market Segments (SAM)



**~35-40
MILLION PONDS
WORLDWIDE**
(estimated)

- Global aquaculture is dominated by pond-based systems

~10-14 MILLION PONDS
suitable for solar floating eco-lights

- ▶ Off-grid / unreliable grid
- ▶ Night feeding / security lighting
- ▶ Smallholder & semi-commercial ponds



Bangladesh - 300K

India - 150K

Vietnam - 75K

Thailand - 50K

Nigeria - 50K

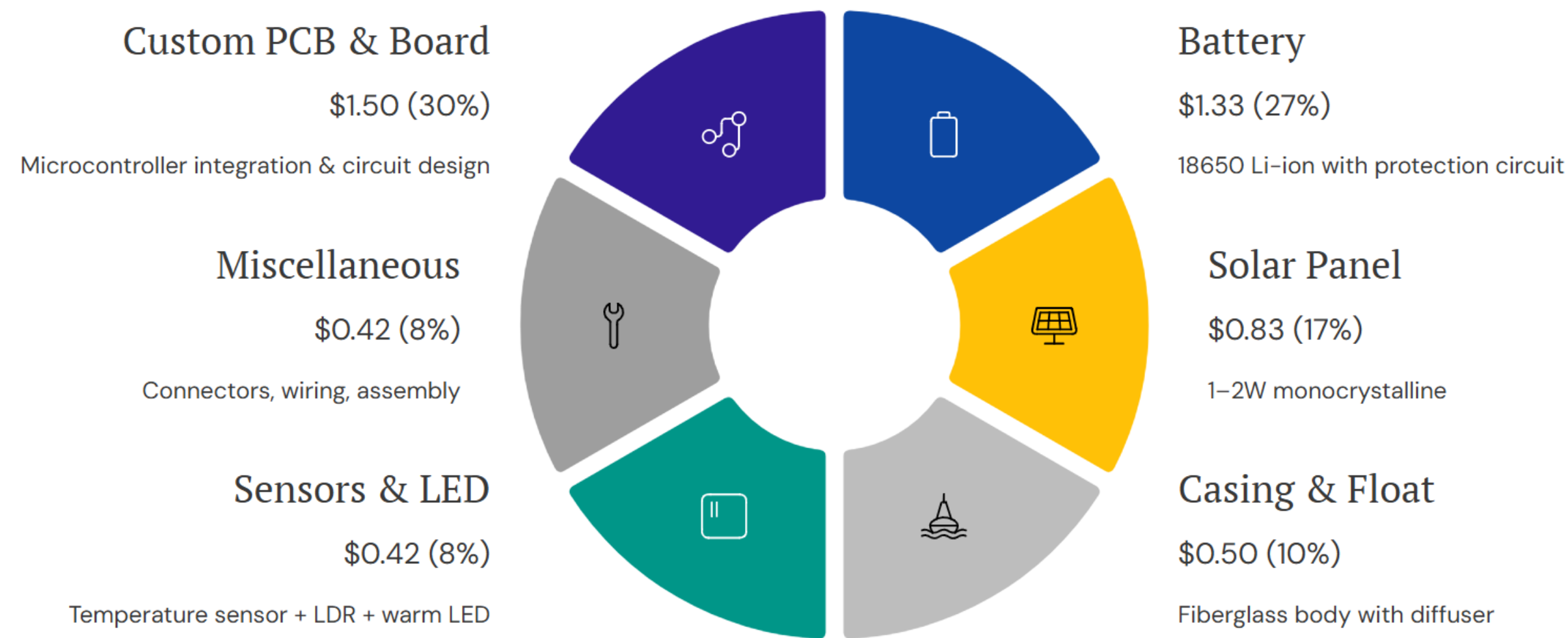
Kenya - 25K

Sources:

- **FAO** – The State of World Fisheries and Aquaculture (SOFIA)
- **WorldFish** – Small-Scale Aquaculture in Asia & Africa
- **IRENA** – Renewable Energy Applications in Aquaculture

* Pond numbers are estimated based on regional aquaculture dominance and smallholder system prevalence.

Cost & Component Breakdown (1000+ Production)



Total System Cost: \$5.00 USD | Low-Cost, Sustainable, Hardware-Efficient Design | Economies of scale: 1000+ unit production

Business Model

Aaloo Kit



9\$

App + Web
Subscription



2\$

Per Month

Competitor



VS



No feeding innovation
No insect-based feeding

JOIN US !

Contribute to Innovation



Visit website: [Aaloo](#)

1 NO
POVERTY



Supports 18M livelihoods,
reduces energy costs

7 AFFORDABLE AND
CLEAN ENERGY



100% solar lighting → 40%
energy cost reduction

8 DECENT WORK AND
ECONOMIC GROWTH



Increases efficiency,
reduces labor, boosts
productivity

IoT-enabled lamps →
scalable & smart
aquaculture tech

9 INDUSTRY, INNOVATION
AND INFRASTRUCTURE



Circular design using
recycled PET bottles

12 RESPONSIBLE
CONSUMPTION
AND PRODUCTION



Promotes healthy aquatic
ecosystems, safe natural
feeding cycles

14 LIFE
BELOW WATER





Team Jealoo

Thank You !